

Institute of Actuaries of India

Subject SA1 – Health and Care Insurance

April 2016 Examinations

INDICATIVE SOLUTIONS

General Comments

Candidates appearing for SA1 subject are expected not only to have the knowledge about the subject but also the ability to apply the knowledge and use judgment.

Candidates who wrote generic answers and didn't relate with the underlying question didn't gain marks.

Candidates who approached the questions, especially the more substantial elements of each question, in a methodical and detailed manner were far more likely to satisfy the examiners and receive a pass in the subject. The mark allocation for each question part gives an indication of the relative length of answer or number of points to be made to gain full marks.

Comments on individual questions

Question 1

- (i) This was generally well answered although several answers failed to mention the most important reason for self-insurance is the flexibility to design benefits and managing cash flows and contributions.*
- (ii) Some candidates did not always appreciate that volatility doesn't always mean poor claims experience. A change in age mix, an epidemic or pandemic or lower could result in uncertainty in the overall experience and hence contribution. The possible mitigation techniques were mostly well answered.*
- (iii) While candidates suggested the types of insurance products (Group PMI, Stop Loss etc.) that can be offered, only successful candidates explained the structure of the cover. Students who explained only the features or benefits of each type of insurance cover didn't gain full marks. Many candidates didn't appreciate that the objective of the Finance Director is to reduce the volatility levels of the contributions.*
- (iv) This was poorly answered. Several candidates seemed to lack awareness of the stop loss cover. Many candidates failed to describe the factors that the actuary needs to consider to decide whether to offer a quote.*

Question 2.

- (i) The part was relatively well answered.*
- (ii) Few successful candidates listed the risks of launching a new product. Candidates who listed all possible risks didn't gain any marks.*
- (iii) Many candidates gave generic answers about product design, pricing, underwriting and claims. Candidates who were able to relate their points with the underlying question gained more marks. The premium reviewability part was poorly answered.*
- (iv) Candidates failed to appreciate the proposed product design and additional risks that will need to be considered to evaluate the proposal. Candidates who wrote generic points without relating with the underlying question didn't gain marks. Also*

candidates who repeated the same considerations as the previous question were not awarded any marks. Overall this question was poorly answered by many students.

Question 3:

- (i) This question part was very easy and well answered.*
- (ii) Candidates failed to appreciate that the question was about the additional requirements should the company start writing health insurance business. Again, generic answers were not awarded any marks.*

Question 4

The question discussed the factors to be considered by the actuary to evaluate the proposal of adding a new rating factor. Candidates who made a recommendation and justified it appropriately were far more likely to gain full marks than candidates who wrote a generic answer regarding product assessment.

Also, candidates who evaluated the proposal from the perspectives of all stakeholders gained more marks. Most candidates failed to describe in sufficient detail and the answers lacked breadth in terms of the aspects to be considered.

Solution 1:**i) Why self-insurance?**

Self-funded health insurance plans can give organizations

1. **Greater degree of flexibility** over the plan's design – The employer can create a health plan that best meets the employee's requirements. Such flexibility and customisation may be difficult to get from insurers.
2. **Better cash flow**- the employer essentially pays claims out-of-pocket from collected premiums / contributions rather than using predetermined premiums to compensate an insurance provider for assuming financial risk.
3. **Reduced administration cost** - While some large organizations might be able to handle the plan internally, most use a third-party administrator to oversee the plan. How much responsibility the employer and administrator each bear will vary. This can allow employers the flexibility of a self-funded plan without assuming management responsibilities with which an organization lacks familiarity
4. The employer can set the contribution rates based on their claims history and adjust the plan in other ways to cut costs. If claims are lower than anticipated, the employer can invest the savings and earn interest.
5. Flexibility and more options around investments – an insurance company's investment policy will be more regulated. However given the short-term nature of the health insurance liability, the investment policy may not be a big differentiator in choosing self-insurance.
6. Elimination of the insurance carrier's risk charge. The employer wouldn't pay for insurer's expenses, profit and cost of capital if it decides to self-insure. If the experience is favourable then the employer would save on those costs. The additional money can be used to enhance benefits.
7. Possible tax benefits. The employer can set up a trust to manage the health insurance and other health related benefits. This may be tax advantageous for the employer.
8. As the group size is large, the claim experience is expected to be stable.

[4]

ii) Reasons of volatility

The volatility in the contribution levels could be due to volatility in the expected claims outgo. The expected claims outgo may be volatile due to:

1. Volatility in incidence rates or
2. Volatility in claim amounts or both.

Volatility in Incidence rates: The incidence rate could be volatile because of the

1. Change in Composition of group – i.e. age / gender / geographical location mix-
 - a. A major merger / acquisition / sale of business could lead to change in the composition of the group.
 - b. Changes in economic cycle may have forced the employer to restructure business and hence the composition of group may have changed as a result.

A sudden change in the composition of group is likely to impact the incidence rates.

2. The average age of the portfolio may be increasing which could increase the volatility in incidence rates.
3. Impact of the wellness or preventive care initiatives - e.g. free cancer screening camp. Such initiative could result into a sudden surge of claims
4. Accidents / pandemic / catastrophe events – The concentration risk is very high within a group. Any catastrophe or pandemic can severely impact the overall claims experience.

Volatility in Claim amounts – The volatility in the claims amount could be due to change in

1. Medical inflation - higher than anticipated medical inflation
2. Medical and technological advancements – newer diseases or treatments may lead to unexpected claim cost.
3. Provider behavior – the employer has very little control over the provider services and billing behavior and hence any change in treatment or billing protocol would impact the claim cost.
4. Change in utilization (Over / Under) pattern between the 3 different benefit types
5. Claims handling and management – claim management could be lenient leading to claims being paid that would have otherwise got denied.
6. Large claims – since the benefits are unlimited, periodic large claims would add to the volatility.

Mitigation

The following mitigation techniques are available to the finance director:

1. Insurance: Insuring some or all of the benefits would ensure that the company pays a fixed premium in a given year to transfer the volatility risk to an insurer
2. Change in benefit structure: The company can re-design the benefit structure by introducing
 - a. Benefit ceilings – providing benefit caps will ensure the maximum liability per event is limited.
 - b. Coinsurance – by introducing cost sharing techniques like coinsurance / deductible, the overall claim cost will reduce
3. Change in eligibility criteria- the employer can introduce a stricter minimum eligibility criteria [e.g. cover only for permanent employees]
4. Stringent claims management will ensure that only the eligible claims are paid.
5. More focus on wellness and preventive care is likely to reduce large claims if the condition / diseases are diagnosed at an early stage.
6. Efficient network management with providers, fixing cost of the treatments and encouraging employees to use the network providers may help in containing outgo.

[6]

iii) Forms and structure of cover

The first step will be to decide which benefits should be covered under insurance. Given the predictable nature of the expected cost of the wellness and the preventive care benefits, it may be reasonable to keep these benefits out of the scope of insurance.

An efficient primary care [i.e. outpatient consultations, medicines] benefit helps in early diagnosis which may help in reducing the incidence and cost of hospitalisation. So including it in the insurance may reduce the overall premium provided the insurer's product offers out-patient cover.

The next step will be to decide the structure of the insurance cover. A group PMI policy may offer most of the benefits currently offered by the employer. However, the policy may not offer unlimited benefits or the cost of offering such benefit may be prohibitive. Additionally, the policy may only pay for acute covers and the company will still need to provide for the employees with chronic ailments who are currently receiving benefit under the existing schemes. There may be other requirements (waiting period, pre-existing conditions etc.) imposed by the insurer which may restrict the ability of the employer to offer to provide comprehensive benefits.

Instead, the employer may consider taking a stop loss cover. Given that both the timing and amount of claims are unknown in advance in the indemnity covers, non-proportional cover will be more appropriate to minimise the volatility.

Non-proportional cover includes individual / aggregate excess of loss and stop loss. Specific stop loss coverage protects the employer against the severity of large losses per individual claimant. Aggregate stop loss coverage protects the employer against abnormal frequency of claims in total rather than abnormal severity of a single claim.

Excess of loss can operate on each risk, on an aggregation of claims, or on a catastrophe basis. It limits an employer's liability to a set Rupee figure per individual, per policy year. This limit is known as a pooling point or deductible or threshold. For example, an employer may elect a pooling point of INR 100,000 per policy year. That means their maximum liability per person on their benefit plan for that policy year will be INR 100,000. The insurer will reimburse for claims in excess of that amount. This is particularly important given the unlimited benefit provided by the employer.

Employers choosing a specific deductible should consider expected severity and frequency of "large" claims for their size and for their risk tolerance. A typical employer's capital resources are more limited than those of an insurance company and therefore, it's reasonable to have lower specific deductibles.

Stop loss covers loss ratios between limits for a whole account over a certain period. The employer's maximum liability is expressed in terms of a percentage of total expected claims, say 125 percent. If paid claims exceed the expected claims by more than 25 percent, the Stop Loss coverage will reimburse the employer for the difference.

The company can also choose a combination of individual excess of loss cover and stop loss cover to limit the overall liability.

For deciding the structure of the cover, one of the main inputs will be to decide the threshold level.

The required threshold level will be such that it meets the company's objective. The finance director's objective is to reduce the uncertainty in the claims experience so the threshold level should be such that the volatility of claims are minimised.

The benefit structure and claims history must be considered to decide a prospective distribution of the loss ratio of scheme. This can be done by plotting the past claims history and statistically fit a loss distribution.

Once the distribution is identified, the next step will be to estimate the distribution of next year's loss ratio. One of the ways is by simulating (stochastically) the possible loss scenarios. Using the confidence interval of the resulting loss distribution a threshold can be chosen that reduces the variance of the loss distribution. Multiple scenarios of the threshold levels can be created for the finance director's consideration or to obtain the insurance quotation.

[8]

iv) Information required by insurer's actuary

Following scheme related information may be sought by the actuary:

1. Demographic Information about members and their dependents
2. Past contributions details
3. Current and Past claims details
4. Expected change in membership profile [e.g. if any planned mergers or acquisitions or sale]
5. Investment policy and expected return
6. Benefit structure (including wellness plans) and entitlement
7. Claims management and administration policy
8. Employee enrollment policy
9. Type of industry and nature of duties performed by employees.
10. Geographical spread of the company
11. Stop loss cover required by the company.

The actuary must consider the following aspects before deciding whether to offer the quote:

The actuary must decide on the basis that the quote meets the company's risk management framework and there is a reasonable chance of meeting the expected profitability targets.

Following aspects will assist in the decision making:

Whether the claims fluctuate considerably from year to year and whether the fluctuation comes only from the claims side and is of a random nature.

The nature of liability must be short-tailed so that it's easier to project the tail risk.

How the retention / threshold level is fixed? Will the ceding company make a loss before the cover comes into effect? If no, then the company mayn't effectively manage claims and the insurer may end up making loss.

The insurer may choose not to pay 100% but 90 or 80% of the stop-loss claims. This may stimulate the ceding company to maintain the quality of its business.

The actuary must consider the risk appetite of its own company and whether the company has sufficient experience in writing such business. He / She may also want to explore if the insurer can get reinsurance coverage and if yes then what terms are being offered by the insurer.

The actuary must check if there are other competing insurers quoting for the business and if yes then what terms they are offering? Are the term profitable or loss making as per his estimates?

Given company's size, the actuary must consider the available capital and impact of the scheme on company's solvency levels if the scheme were to be insured. The impact of pandemic and catastrophic events should also be considered.

[7]

[25 Marks]

Solution 2:

i) Why should the company launch the severity based CI product?

1. A severity based CI product is more comprehensive than standalone CI as multiple claims are possible which will enhance policyholders' satisfaction and retention.
2. As a variant of standard product it will help the insurer to differentiate its offerings from its competitors and the new product may be more profitable than existing products
3. There is a desire to grow the business or to compete with other insurers who are currently writing severity based CI.
4. There may be a demand for this product in recent years combined with declining sales for traditional health indemnity products or standalone CI products.
5. The company may want to explore cross-sell and upsell opportunities with the existing policyholders. The product may be designed to integrate with the existing covers so that cost overlaps may be extracted and the customer may pay a lower premium for enhanced benefits
6. the demand may be coming from the distribution channel

7. The company may be encouraged to write complex CI by reinsurers. The reinsurer may be helping in designing the product and its benefits
8. To make efficient use of spare capital. [5]

ii) Risks of a new product launch?

1. Product substitution risk- the new product may outsell an existing profitable product.
2. Too many sales - The product if becomes too successful and is a capital intensive product then it may end up using up capital required elsewhere and may destabilise the sales across the product range.
3. Sales failure – poor sales would mean that the development cost and other sales related cost may not be recovered. At the same time, the morbidity experience for a smaller portfolio may be poor and more volatile.
4. Expense overruns – the product may be very expensive to sell and the total cost of sales may results in significant cost overruns resulting in reducing the available capital
5. Product design failure – the product doesn't meet customers' requirements or the policy conditions are poorly worded leading to claims not priced for or litigations.
6. Product costing failure [parameter error] – the product has been under or overpriced due to lack of relevant data, error in modelling or inadequate underwriting of risk. [5]

iii)

a) Product design and Pricing Considerations

The main points to consider are

Product Design

The product design should aim to meet customer's requirements without compromising on company's risk management and balancing the shareholder's risk-return requirements.

The first step is to choose the illnesses to be covered under the product. The main consideration are:

- Expected frequency - a relatively rare occurring illness may not have sufficient data to be accurately priced. At the same time, a commonly occurring illness may not be seen as critical.
- Cost associated with treatment and impact on future lifestyle- the treatment of the illnesses leads to consideration change in the lifestyle of the individual.
- The common perception [scare] about the illness in the minds of the customers

A market research may be conducted to check if similar products are available in the domestic market. If yes then what benefits do they offer? What's the market response for such products? Can the insurer's product offerings be differentiated?

The next step will be to define the severity level of each of the illnesses. The severity levels must reflect the expected financial loss [both due to treatment and loss of income] on occurrence of that illness.

The definitions will also need to be decided. If the CI conditions covered are different from the regulatory approved definition then the definitions will need to be decided. A balance needs to be struck whether the definitions are stringent or liberal.

A very stringent definition means few claims will be payable and may lead to repudiations if claims are filed and definitions are not fully met. This may create reputation risk.

A liberal definition may lead to many claims being payable which may not strictly meet the criteria. So a balance approach needs to be decided.

The company may seek reinsurer's assistance for design and pricing.

The product boundary conditions are to be set such as

- a) Entry / exit ages,
- b) Minimum / maximum sum assured.
- c) Premium and policy term
- d) Distribution channel
- e) The reviewability of the premiums. A product with guaranteed premium may make the product un-competitive or unaffordable. A higher premium may attract unhealthy lives or may lead to anti-selection.
- f) Waiting period and survival period after claim
- g) Cooling period between two successive claims- this will be very important consideration for this product.
- h) Waiver of premium post the first claim may be considered
- i) General Exclusions

Pricing

The pricing actuary needs to estimate the incidence rates for the conditions covered. The incidence rates will need to be split by first and subsequent occurrences and also the interlinkages between various conditions covered. Depending on the conditions / illnesses covered the actuary will need to decide the inter-dependency of the conditions.

The company has been selling only standalone CI products so the existing experience mayn't be directly relevant since the claims paid information will be available only for the claims that met the definition. However, under the new product, the benefit will be payable based on the severity of the illness and that mayn't have been captured. Moreover the current product may be paying only on first occurrence of a given illness.

The company may be able to use the experience under the indemnity health plans if the claims have been properly classified using the International Classification of Diseases or ICD diagnosis and procedure codes.

Another potential source of the data could be hospital discharge summary where the cause of hospitalization and resulting treatments are properly captured. This may be hard to get access to or the cost of acquisition of the data may be very high. Also the corresponding exposures may be hard to estimate given that there may be more than one hospital in that locality or it will only have the cases where the patient has got admitted.

Given the uncertainty over the availability of credible data, the actuary may be comfortable in offering annually reviewable premiums.

The company can also approach the reinsurers for a quota share based quotation and adjust the premium rates to estimate the underlying incidence rates.

Few other aspects to be consider are

- The frequency of claims under this product will be higher the standalone CI but the severity may be less as claims may be reported early.
- The cost of claim assessment will increase given the possibility of multiple claims and hence it needs to be factored into pricing.
- The company may want to spend on advertising and marketing given the new product concept is slightly more complex and may need more training for the advisors / distributors. The cost of launch is likely to be high.
- The investment strategy, though regulated, may not be very different than given the short tailed nature of product.
- The capital requirements under this product will be higher than the traditional CI product. The availability of sufficient capital must be checked to write the expected volume
- The risk discount rate to discount the future profit cash flows may need to be adjusted to reflect the un-certainty in the best estimate incidence rates.
- Another key assumption will be the expected sales volume. Research must be carried out to estimate the demand of the product.
- The mortality rates assumption for this product will also need to take into account the post illness mortality.
- Finally the profit margin is likely to be higher given the levels of uncertainty involved.

Once a set of premium rates are generated it needs to be tested for profitability, competitiveness and sensitivities to ensure that the product remains viable under various adverse scenarios and attractive for customers.

b) Underwriting and Claims

Underwriting

The main objective of underwriting will to be select lives as assumed in the pricing basis.

The underwriting approach for this product will be broadly similar to the underwriting of the existing standalone CI product provided the boundary conditions are similar.

The key difference between the two products is that the expectation of an early claims under this product. Hence, the underwriting questionnaire must be focused to reduce the risk of anti-selection. The initial waiting period under the product may also help in reducing early claims and hence the length of the waiting period in the new product is likely to impact the underwriting approach.

The underwriting questionnaire and approach will also need to take into account the conditions covered by the product. For e.g. if the product offers cover against cancer then a questionnaire related to family history of cancer should be asked.

Similarly, the trigger for medical underwriting must be clearly specified and the medical underwriting shall be influenced by the type of illnesses covered under the product.

The company may want to compare the underwriting practices for similar products offered by competition.

The overarching principle is that the cost of underwriting must not exceed the expected benefits derived from it.

The company's board approved underwriting philosophy will also act as a guiding principle for setting the underwriting approach. Also it should take into account any regulatory guidelines or restrictions around underwriting [e.g. genetic testing etc.]

The guidelines must indicate how to treat applicants with pre-existing conditions. The guidelines must indicate which conditions to be covered and when. Also, the additional risk of covering pre-existing condition must be priced adequately.

Claims

The claims are expected to occur earlier than the current standalone CI product. The claim assessors need to be trained to be able to categorize the claim as per the severity.

The claims manual or guidelines need to be reviewed or updated, if necessary, to reflect the new claim triggers.

The claims assessor will need to investigate each claim to ensure the eligibility. The data entry and amount of information required to assess a claim will significantly increase for this product and hence the time taken to assess the claims is likely to increase and so is the cost. Is the current team strength sufficient? The existing team may need additional training to assess claims under this product.

The claim assessor may also need more specific evidences to ascertain the severity which the treating doctor may not have considered at the time of treatment. This may lead to delays in claims processing and may lead to rejection in claims. Hence, it is important to clearly state the requirements in the policy documents to avoid any issues at the claims stage.

There's likely to be difference in opinion between the treating doctor and the claim assessor especially when the claim assessor decides to downgrades the category of the claim than the one assigned by the doctor. The benefit payable will then reduce and it may lead to litigations. Increase repudiation may lead to reputational risk.

c) Premium Reviewability

Following aspects should be considered while offering reviewable premiums:

- The company is offering this product for the first time, so it doesn't have credible experience and hence confidence in its pricing. Offering reviewable premium will allow the insurer to adjust its pricing basis as it gains more experience.
- This may also help in offering lower premiums as cost of guarantee may be very high for this product given the lack of past experience.
- However, the company must consider the impact of offering reviewable premiums on the marketability of the product especially if a similar product is being sold in the market with guaranteed premiums.
- Moreover there is a huge reputational risk if the company has to increase the premium due to poor experience. So the insurer should base reviewable premiums on assumptions that they believe are valid for the full term of the policy and the assumptions on which premiums are based should be reviewed regularly.
- The premium reviewability clause should be prominently shown in product literature and the basis for reviews should be clearly set out.
- The customers should be explained clearly what premium reviewability means and under what circumstances the premiums may change. It should be clearly stated that the premiums change will be based on the portfolio claims experience and not individual's claim experience.
- The benefit of offering non-guaranteed premiums should be passed on to the customer in terms of lower premiums.
- The insurer must comply with the regulations around offering reviewable premiums.

d) Sales and Other consideration

Sales

The product and benefit structure will be very complicated, so the chances of mis-selling is likely to be higher than the current product. Hence the company needs to carefully choose the distribution channels that will sell this product.

If the company uses multiple distribution channels to sell the existing CI product then a quick analysis of the performance [in terms of lapses, early claims, and early cancellation] of each of the distribution channels will reflect the efficiency of the channels.

The regulatory aspects around selling the product will also need to be considered. E.g. the benefit illustration must be clear and unambiguous. The key feature document should

explain all benefits available under the product. Special mention around benefits post first claim will need to be explained.

The company may consider training individual agents specially to sell this product. They may choose the agents based on their existing performance in terms of selling complicated products and further train them to sell this product. Another option is to have a dedicated sales force specially trained to sell this product.

The advisor will need to relatively spend more time in explaining the product features and policy conditions than traditional CI product. Hence they may expect a higher remuneration. Overall the cost of selling this product may be higher compared to the existing product.

Other channels that may be explored are

1. Internet based selling or web aggregators
2. Direct selling via telephone, emails, newspaper advertisement with a toll free number]
3. Dedicated sales force
4. Bancassurance if available

Other Considerations

The administration of this product will be more complicated since there will be multiple claims.

At any given time there will be policy holders who may have never claimed, policy holders who have claimed once or more than once and then who have either died or lapsed their policy. This needs to be considered while designing the administration system.

The reserving approach will also have to be decided. A multiple state modelling may be required to segregate the lives depending on their health status and their transitional probabilities will need to be estimated.

[20]

iv) Multi-pay CI with choice to customer

From the perspective of the customer, the option to get more choice is likely to increase the attractiveness of the product. The customer will be able to choose the cover and respective sums assured that best meet his/ her requirements.

If the customer already has existing cover and wants additional cover only for certain illnesses then this product will easily integrate with the existing cover and the customer will only pay for the cover he/ she requires.

However, it may also increase the risk of poor selection and may lead to insufficient or poor coverage. The customer may not be able to choose the best option for himself/ herself.

For the insurer, the product concept is novel hence it will give the insurer the first mover advantage and will help in increasing premium and profitability of the company.

Having said that the insurer must also evaluate the following aspects:

The choices is likely to increase the chances of anti-selection against the insurer. The customer is likely to choose the baskets where he/ she believes the chances of making claim is higher than being reflected in the premium. The anti-selection risk may need to be factored in pricing and managed through underwriting.

The illnesses will need to be clubbed for each of the buckets in a way that the clubbed conditions are interconnected while the buckets are independent. This may require thorough medical research. The insurer may require external help from medical consultants or reinsurers in designing appropriate buckets.

The pricing of the product will have to be dynamic. The pricing actuary needs to decide whether the buckets to be priced independently or in combinations. There are 7 possible combinations in the proposed design. The premium tables will need to be calculated for each of the combinations. Also, the policy becomes paid up after the first claim, so a waiver of premium rider will have to be inbuilt into the premium.

The incidence rates for each illness will have two components. First will be the first ever occurrence and then condition probability of the same illness post the occurrence of another illness from separate bucket.

The portfolio distribution would need additional assumption of proportion of customer opting for either one bucket or combination of buckets. Also the sum assured for each of the buckets could vary. This will add to the portfolio risk. .

The premiums are to be guaranteed for 5 policy years. The cost of guarantee should reflect the additional risk of increase in incidence rates due to medical advancements. Offering guarantee may also increase the capital requirements to support additional reserves. The sufficiency of capital will need to be assessed.

The company may want to check with the reinsurers if they are willing to support the product and the indicative premiums.

The underwriting approach for this product may vary depending on the type and number of buckets chosen by the customer. I.e. the underwriting questionnaire and medical underwriting triggers may depend on whether the customer choosing the cancer bucket or cardiovascular bucket.

This will make the underwriting very complex and expensive to administer.

Another aspect will be sale of this product. Given the new concept, the company may need to spend on marketing the product and at the same time train the distribution channel. The product may be easier to explain than the severity based CI. However it may still require a well-trained distribution to sell this product.

This administration of this product will be very complex. The policy admin system should be able to handle the additional complexity due to choice of buckets and to deal with the paid up policies with more than one claim is yet to be incurred.

[10]

[40 Marks]

Note: the student is expected to evaluate the proposal from the perspective of:

1. *Customer*
2. *Pricing Actuary*
3. *Underwriting*
4. *Sales*
5. *Administration*
6. *Shareholders*

The answer to this question should not be a repeat of question 2.3.

Solution 3:

i) Advantages & Disadvantages of moving into Health Insurance business

Advantages:

- High growth rate associated with this line helps to build volumes/top lines
- Relative ease of sale as compared to other non-life products (pull rather than push)
- Government/ political support in favour of promoting this line (by introducing increased tax incentives) to meet emerging population demand
- Non-conventional channels (online/ digital/ Direct marketing or Telemarketing) can be used to promote and distribute health insurance product
- Helps to attain overall business profitability due to low loss ratios (for retail customers) attached with health insurance
- Space for lots of innovation (in terms of products and features) to meet emerging need and capture untapped market share
- Helps to build customer base which could be used for cross-sell and upsell to the existing customers.
- The regulatory requirements pertaining to estimation of IBNR, UPR, capital and solvency assessment is similar, it will be relatively easier for the company to manage.
- The company could make efficient use of spare capital.

Disadvantages:

- Emerging risk (ailments/ illnesses/ treatments) and demand leads to high uncertainty with respect to future experience
- High risk of fraud and anti-selection as compared to other lines of business
- High level of competition with advent of standalone health insurance companies in industry
- Lack of relevant experience and infrastructure available in-house would result in high operational expenses which could cause high initial capital strain
- Regulatory risk coming from unexpected changes in health regulations which influences strategy and planning attached to this portfolio
- Reputation risk due to sensitive nature of this line of business (where customer avails benefit in the time of medical distress) and hence more prone to grievances which could affect sale of other lines of business
- Competition from niche player may not result in expected market volumes.
- Decision to pull out from this line business in future could adversely impact company's reputation and other lines of businesses.

[8]

ii) a) Asset Liability Management

The company can utilise existing ALM model (in terms of form and format) for modelling assets and liabilities attached to health insurance business.

Liability modelling:

- In order to understand the underlying behaviour of claims, the lag table/ delay table for emerging claims development needs to be estimated. The claims development pattern could be estimated by using information available in public domain (e.g. IIB information and public disclosures of company selling similar products)
- If the company plans to cover overseas market as well (like in travel product), the exchange rate impact of currency also needs to be factored in while projecting claims development.

Asset Modelling:

- Given the short tail of health insurance business, the assets (like short term bonds, money market instruments, .etc.) could be considered to match the underlying claims experience.
- The admissibility of the assets as per IRDA regulations need to be considered while selecting them or allocating them to health insurance business

Scenario Testing:

- The term/ durational matching of the assets and liabilities should be testing under key scenarios, so as to ensure adequacies in most of the possible future scenarios
- The key scenarios like +/- 20% loss ratios, +/- 2% interest rate, +/- 20% Business volumes, etc. can be tested for the purpose.
- Simulations could be run to identify scenarios where mismatches would be beyond target range (e.g. +/-5% difference)
- The scenarios (like large losses, Reinsurers payment default) where matching cannot be achieved to the desired level should be covered by keeping aside reserves for the purpose.

b) Solvency/Capital assessment

In order to assess capital requirement the Company needs to build a model which takes into account various factors that are discussed below. As the Company is established General insurer with experience in various lines of insurance other than Health it should have some internal models in place that can be modified and the value of parameters can be estimated using internal benchmarks coupled with what is available in public domain.

Following are the factors that may have impact on the final output which is amount of capital required over next few years:

- The first one is the quantum of premium it wants to achieve in next few years. This should flow from the overall business plan prepared for next couple of years and should be in line with the aspirations of the management and Board.
- Next comes the channels through which it wants to market its products. This will have influence on the amount of intermediation it needs to pay out. The synergy and potential conflict with the existing channels will be an important consideration for this factor.
- The Health line of business can be further divided into mainly three sub-lines, viz., Retail, Corporate and Govt. Schemes. Determining contribution from each of the line will be important consideration as the experience in terms of loss ratio, expenses, etc. will vary widely in these lines.
- The estimation of loss ratio is one of the important input for the model. In Health LOB loss ratio is a function of various factors including level of underwriting and claims. The industry statistics suitably modified can be the starting point for this. It is equally important to estimate loss ratio for new and renewal cohorts separately and consider important factors like medical inflation while forecasting.

- Renewal rate also is required to be estimated for projecting GWP as well as loss ratio for a no. of years in future.
- Expenses should be estimated related to manpower, general administration, claims related, etc. These should be categorised in fixed and variable. Here internal data modified with change in LOB can be of good use.
- There are nuances related to Health LOB that should be specifically dealt with, like it is a short-tailed line of business and therefore the float available for investment can be lower as compared to other LOB's other things remaining the same.
- The required rate of return on capital invested will also determine the assessment of capital. This must already be available but it might be that shareholder may settle with lower return considering the low volatility in health as compared to other LOB's (lower risk margin). The impact of inclusion of this LOB should also be studied in terms of overall Solvency Ratio and it should be determined if the solvency ratio is same with and without Health LOB or not. It need not be same but there must be safeguards in place as not to breach regulatory minimum threshold.
- There are a number of other factors like, investment income, reinsurance cost, income tax rate, etc. but are easy to determine.
- For critical factors parameter values can be varied to see the impact on required capital and therefore impact of various risks can be assessed.

The solvency requirements will be as per the existing IRDAI regulations. Since the company is already be calculating for other lines of businesses, it may find it easier to meet the requirements for calculating the solvency capital for health business.

c) Claims management

The health insurance business has high frequency and managing claims efficiently is an important service requirement to build this line of business appropriately.

Health claims are taken at the time of medical distress by an insured and timely and effective delivery of services are far more important (from reputation perspective) for this insurance line than other non-life class of business written by the company.

The claims management could be handled in two ways; outsourcing it to Third Party Administrators at a fee or developing in-house capabilities and infrastructure.

The third party administrators have expertise and capabilities to handle health claims and administration and would save company from immediate capital strain from cost and service management perspective.

The option of developing in-house claims management eco-system is a cost intensive option on immediate basis as it involves setting up internal health claims team, developing health claims system and tying up with network hospitals (for cashless claims).

Though, the in-house claims management has long term benefits in terms of better expense control, reduced fraudulent claims, better claims service, etc.

Both options need to be evaluated with respect to internal capital requirements determined during capital measurement exercise (refer part (b) above) and the cost effective and time efficient option should be selected with reference to long term business objectives of the company. [15]

[23 Marks]

Solution 4:

Proposal Evaluation

Note:

The student is expected to evaluate the proposal from the perspectives of all the stakeholders involved mainly:

- *Consumers*
- *Shareholders*
- *Regulatory*
- *Pricing*
- *Sales & Marketing*
- *Underwriting & Claims*
- *Administration*
-

The indicative answer provided below is intended to list out main aspects that need to be considered for making a recommendation. Candidate making a recommendation and appropriately justifying it should be awarded full marks.

Product Design and Pricing

The proposal will lead to premium differentiation with respect to smoking status of the applicant. The premium for non-smokers is likely to be lower than smokers. This may attract non-smokers given that no other company is currently offering the discount in the market. Hence, it is likely to increase the new business sales and profitability for the company.

A marketing survey may be conducted to establish the need of the product and whether such a proposal would be attractive to the customers.

The Actuary needs to decide the approach for pricing the smoking status. Since, the company doesn't capture smoking status hence, the existing claim experience may not be of much help.

The definition of smokers will need to be decided. E.g. whether an individual smoking more than 10 cigarettes a day or 20 cigarettes should be considered a smoker. Also the duration of smoking may be considered for the definition. Should tobacco chewing be considered as it is equally harmful?

The actuary needs to evaluate the impact of smoking on the frequency and/or severity.

It is a known fact that the adverse impact of smoking increases with the duration of smoking, so the actuary needs to consider if there is advantages of offering the premium differentiation at younger ages. If the premium differentials (between smoker and non-smokers) is insignificant at younger ages, will the product still be attractive?

The actuary may need to seek assistance from the reinsurers or look for reports published by government / industry bodies monitoring the impact of smoking on health and wellbeing. The industry insured data on smoking will not be available.

The actuary may also look at the premium differentials used in markets where smoking status is used as a rating factor and use it with judgement

Since the premiums are yearly renewable, the actuary may be willing to take risk provided there are sufficient margins for adverse deviation.

Sensitivity testing will need to be carried out to test the volatility of the assumptions on the profitability. The cost of any uncertainty will need to be added in the pricing. The capital availability and impact on overall solvency post the launch should be carefully evaluated.

The overall pricing approach will also need to be discussed and agreed with the regulator.

Underwriting, Claims and Administration

The insurer will need to decide how the smoking status will be determined at the time of application.

A declaration based smoking status i.e. where the policy holder declares in the application form about smoking status, will expedite the underwriting process. However, it may be anti-selective given the poor disclosures levels in India. The insurer may also find it difficult to reject a claim on the basis of false declaration. This may lead to disputes and increase the reputation risk.

A medical test (either blood or urine cotinine) will correctly identify the smoking status. However it will be expensive and may delay the application issuance process. The premiums may become uncompetitive for yearly renewable product. Also, the company needs to decide if the underwriting decision remains effective for next 2-3 years. Otherwise repeating the test at every renewal may not be attractive and cost effective.

The actuary may consider if the term of the product could be increased to spread the underwriting cost.

If the company decides to share the cost of medicals with the applicants may impact the attractiveness of the product. This is applicable especially for younger individuals where the premium differentials (between smokers and non-smokers may not be significant).

For older ages, if the total cost (premium + cost of medicals) is cheaper than the current premiums then the policyholders may be willing to undergo medical underwriting.

The positive selection (due to medical underwriting) is likely to further reduce the premiums due to lower expected claims cost.

The underwriting team strength will need to be enhanced since each applicant will be medically screened. This may increase the set up cost for this product.

The application forms, policy documents and administration systems will need to be modified to capture the smoking status.

Sales and Distribution

Given this is a new product, the company may want to market is aggressively to capture the market before the competition introduces a similar product. This is likely to increase the cost of marketing the product.

The distribution channels may also need to be trained to explain the new product. There may be a push back from the distribution if medical underwriting is made compulsory for non-smokers as it will delay the product issuance.

The overall cost of this introducing this product is likely to be high and it can get absorbed if the company is able to sell big volumes of business, so the assumption of expected business volume will be very crucial in the decision of whether to introduce the product. The action of competition will need to be carefully assessed. If the competition launches a similar product shortly then the company may quickly lose the edge and expected volumes will be impacted.

The actuary may consider multiple scenarios of the expected volume to perform a sensitivity test on the overall profitability

Existing business

For the renewal of the existing business, the company will need to decide if the new product will be extended to the existing policy holders at renewal. Since the smoking status was not captured when the policy was originally written, the policyholders may need to undergo fresh underwriting to determine the smoking status.

The insurer will need to decide who will bear the cost of medical underwriting.

Further the premiums for the smokers is likely to be higher than the existing premium. They may either have to pay a higher premium to continue the coverage or accept lower Sum Assured (provided it is offered). This may result in smokers lapsing their policies. Some of

the lapses could be anti-selective as smokers who have been claiming in the past will stay as they may not get cover with another insurers while healthy smokers may move.

This is likely to create a mismatch in Policyholders Reasonable Expectations (PRE) especially for (smokers) policyholders who have been renewing their policies for a long time.

The treatment of existing customers may need to be thoroughly discussed with the regulators and approvals may be required.

[12 marks]
